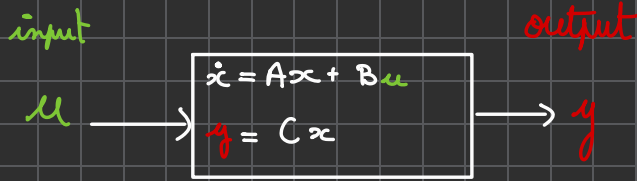




01) Control in state-space.



$x(t)$: state vector (ex: $x = (x_1, x_2, \dots, x_n)^T$)

A : Matrix

B : Matrix or vector

C : Matrix or vector

Example with $n=3$:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad C = (1 \ 0 \ 0)$$

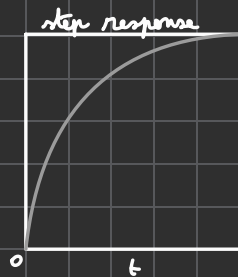
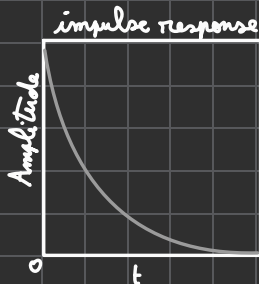
1st order system (transfer function)

$$G(s) = \frac{y(s)}{u(s)} = \frac{2}{s+1}$$

State-space realization of $G(s)$

with $n=1$: $A = -1$, $B = 1$, $C = 2$

$$\begin{array}{l} \dot{x} = Ax + Bu \\ y = Cx \end{array} \rightarrow \begin{array}{l} \dot{x}(t) = -x(t) + u(t) \\ y(t) = 2x(t) \end{array}$$



2nd order system

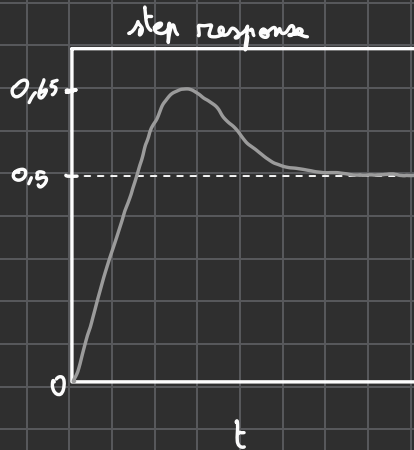
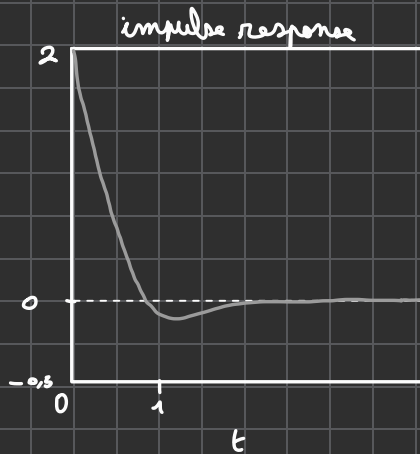
$$G(s) = \frac{2s+1}{s^2+3s+2}$$

State space realization of $G(s)$:

with $n=2$: $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $C = (1 \ 2)$, $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\dot{x} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

$$y = (1 \ 2) x$$



stability? Yes, • impulse response: the system converges to zero.
• step response: the system stabilizes at a constant value of 0.5

overshoot: $D\% = \frac{y_{max} - y_{\infty}}{y_{\infty}} \times 100$ (only for step response)

$$= \frac{0.65 - 0.5}{0.5} \times 100 = 30\%$$

3rd order system

$$G(s) = \frac{2s+1}{s^3+4s^2+3s+2}$$

State space realization of $G(s)$:

$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -3 & -4 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$

$$y = (1 \ 2 \ 0) x$$

How to generate mathematical models of plant?

- ↳ 1) Based on physical laws (E.g., electrical circuit laws)
- ↳ 2) Based on experimental dataset (System identification)

A, B, C, D are given using modeling methods

Differential equation to state-space representation.

$$\ddot{y}(t) + a_1 \dot{y}(t) + a_0 y(t) = u(t)$$

$$\Downarrow \text{State } x(t) = \begin{pmatrix} \dot{y}(t) \\ y(t) \end{pmatrix}$$

$$\begin{aligned} \dot{x} &= \begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u \\ y &= (1 \ 0) x \end{aligned} \quad \left. \vphantom{\begin{aligned} \dot{x} &= \begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u \\ y &= (1 \ 0) x \end{aligned}} \right] \text{controllable canonical form}$$

Transfer function to state-space representation.

$$G(s) = \frac{cs + d}{(s+a)(s+b)}$$

$$\Downarrow$$

$$\begin{aligned} \dot{x} &= \begin{pmatrix} -a & 0 \\ 0 & -b \end{pmatrix} x + \begin{pmatrix} 1 \\ 1 \end{pmatrix} u \\ y &= (\gamma_1 \ \gamma_2) x \\ \gamma_1 &= \frac{-ca + d}{-a + b} \quad \gamma_2 = \frac{-cd + d}{-b + a} \end{aligned} \quad \left. \vphantom{\begin{aligned} \dot{x} &= \begin{pmatrix} -a & 0 \\ 0 & -b \end{pmatrix} x + \begin{pmatrix} 1 \\ 1 \end{pmatrix} u \\ y &= (\gamma_1 \ \gamma_2) x \end{aligned}} \right] \text{diagonal canonical form}$$